Coffee Sales Performance Analysis

Objective

To Analyze coffee sales data to identify top-selling products, customer buying patterns, payment preference,and sales trend over time-enabling, data-driven decisions for marketing inventory, and operations.

In [ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
```

Load Data Set

In [ ]:
```
coffee= pd.read_csv("Coffee.csv")
```

Read First and Buttom 5 rows

In [ ]:
```
coffee.head()
```

Out[ ]:

	date	datetime	cash_type	card	money	coffee_nam
0	2024-03-01	2024-03-01 10:15:50.520	card	ANON-0000-0000-0001	38.7	Latt
1	2024-03-01	2024-03-01 12:19:22.539	card	ANON-0000-0000-0002	38.7	Hc Chocolat
2	2024-03-01	2024-03-01 12:20:18.089	card	ANON-0000-0000-0002	38.7	Hc Chocolat
3	2024-03-01	2024-03-01 13:46:33.006	card	ANON-0000-0000-0003	28.9	American
4	2024-03-01	2024-03-01 13:48:14.626	card	ANON-0000-0000-0004	38.7	Latt

In [ ]:
```
coffee.tail()
```

Out[ ]:

	date	datetime	cash_type	card	money	coffee_r
3631	2025-03-23	2025-03-23 10:34:54.894	card	ANON-0000-0000-1158	35.76	Cappu
3632	2025-03-23	2025-03-23 14:43:37.362	card	ANON-0000-0000-1315	35.76	C
3633	2025-03-23	2025-03-23 14:44:16.864	card	ANON-0000-0000-1315	35.76	C
3634	2025-03-23	2025-03-23 15:47:28.723	card	ANON-0000-0000-1316	25.96	Amer
3635	2025-03-23	2025-03-23 18:11:38.635	card	ANON-0000-0000-1275	35.76	

Shape of Data (Columns & Rows)

In [ ]:
```
coffee.shape
```

Out[ ]:
```
(3636, 6)
```

In [ ]:
```
print(f"The Coffee Dataset have Total Number of Rows: {coffee.shape[0]} and \r
```

```
The Coffee Dataset have Total Number of Rows: 3636 and
 Total Numbers of Columns 6
```

Cleaning and Preparation

In [ ]:
```
#information of coffee dataset
coffee.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3636 entries, 0 to 3635
Data columns (total 6 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   date         3636 non-null   object 
 1   datetime     3636 non-null   object 
 2   cash_type    3636 non-null   object 
 3   card         3547 non-null   object 
 4   money        3636 non-null   float64
 5   coffee_name  3636 non-null   object 
dtypes: float64(1), object(5)
memory usage: 170.6+ KB
```

Check missing & duplicate

In [ ]:
```
#mising data
coffee.isnull().sum()
```

Out[ ]:

	0
date	0
datetime	0
cash_type	0
card	89
money	0
coffee_name	0

dtype: int64

In [ ]:

```
#check percentage-wise
round((coffee.isnull().sum().sum())/(coffee.shape[0]*coffee.shape[1])*100,2)
```

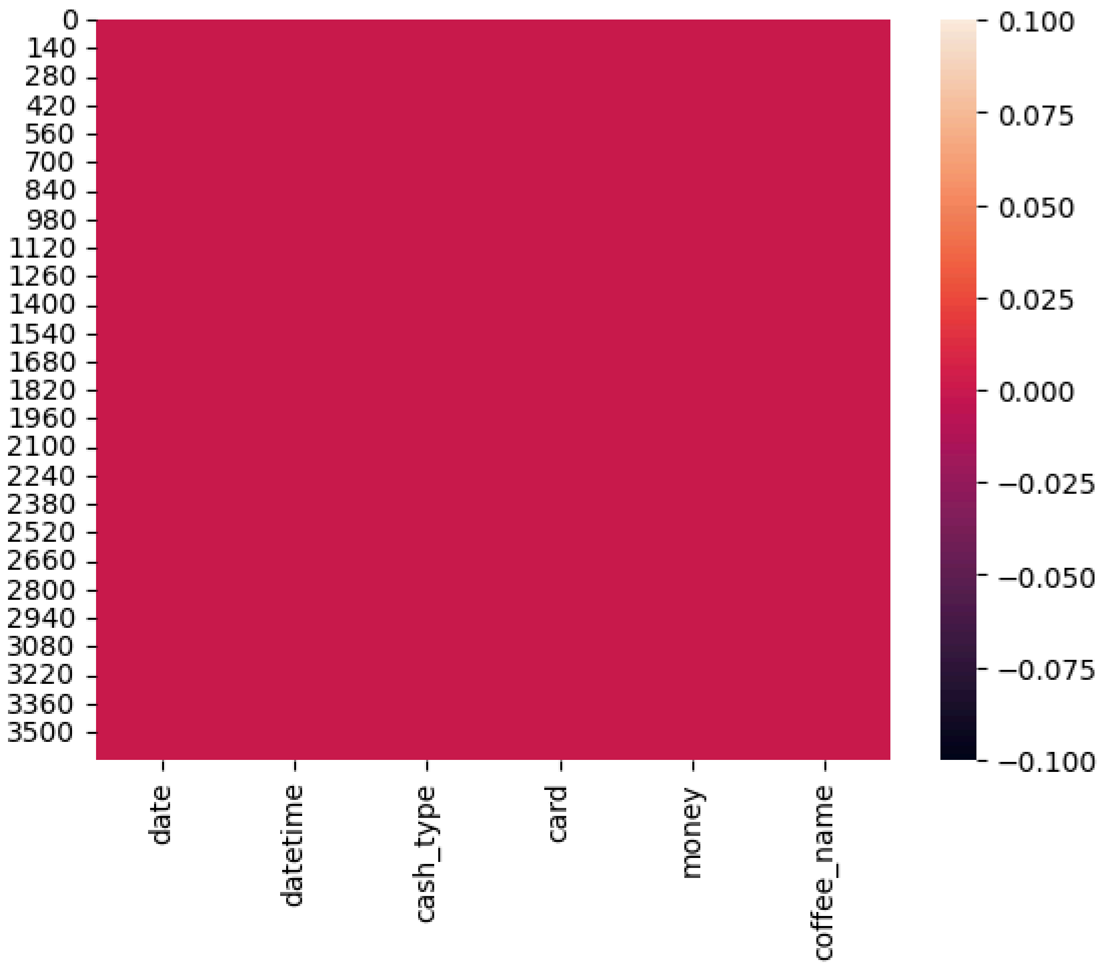
Out[ ]: np.float64(0.41)

In [ ]:

```
# handling missing
coffee["card"].fillna("Unknown", inplace= True)
```

In [ ]:

```
sns.heatmap(coffee.isnull())
plt.show()
```



In [ ]:
```
#check duplicate
coffee.duplicated().sum()
```

Out[ ]:
```
np.int64(0)
```

Convert datetime & drop date columns

In [ ]:
```
coffee["datetime"] = pd.to_datetime(coffee["datetime"])
```

In [ ]:
```
coffee.drop("date", axis =1, inplace=True)
```

Check Column Type

In [ ]:
```
coffee.dtypes
```

Out[ ]:

	0
datetime	datetime64[ns]
cash_type	object
card	object
money	float64
coffee_name	object

dtype: object

Create Derived Column (Months , Weeks & hours)

In [ ]:
```
coffee["Month"] = coffee["datetime"].dt.month_name().str[:3]
```

In [ ]:
```
coffee["Week"] = coffee["datetime"].dt.day_name().str[:3]
```

In [ ]:
```
coffee["Hours"] = coffee["datetime"].dt.hour
```

Exploraty Data Analysis & Visualization

In [ ]:
```
coffee.describe()
```

Out[ ]:

	datetime	money	Hours
count	3636	3636.000000	3636.000000
mean	2024-10-01 02:35:30.535053568	31.746859	14.166667
min	2024-03-01 10:15:50.520000	18.120000	6.000000
25%	2024-07-03 16:54:06.084750080	27.920000	10.000000
50%	2024-10-07 02:55:12.649500160	32.820000	14.000000
75%	2025-01-08 07:55:20.299750144	35.760000	18.000000
max	2025-03-23 18:11:38.635000	40.000000	22.000000
std	NaN	4.919926	4.227603

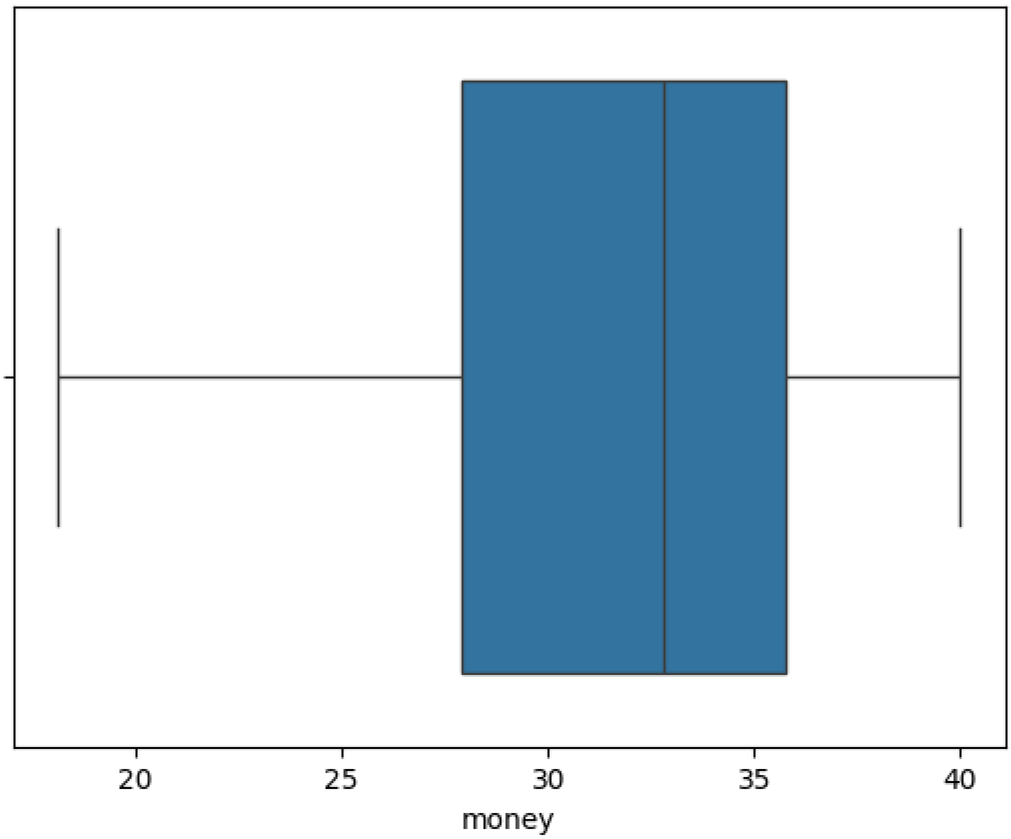
In [ ]:
```
coffee.describe(include="object")
```

Out[ ]:

	cash_type	card	coffee_name	Month	Week
count	3636	3636	3636	3636	3636
unique	2	1317	8	12	7
top	card	ANON-0000-0000-0012	Americano with Milk	Mar	Tue
freq	3547	129	824	525	585

In [ ]:

```
#check Outlier
sns.boxplot(x ="money", data =coffee)
plt.show()
```



The ‘money’ column shows consistent values with no outliers detected.

Q1: Top-Selling Coffee Products

In [ ]:

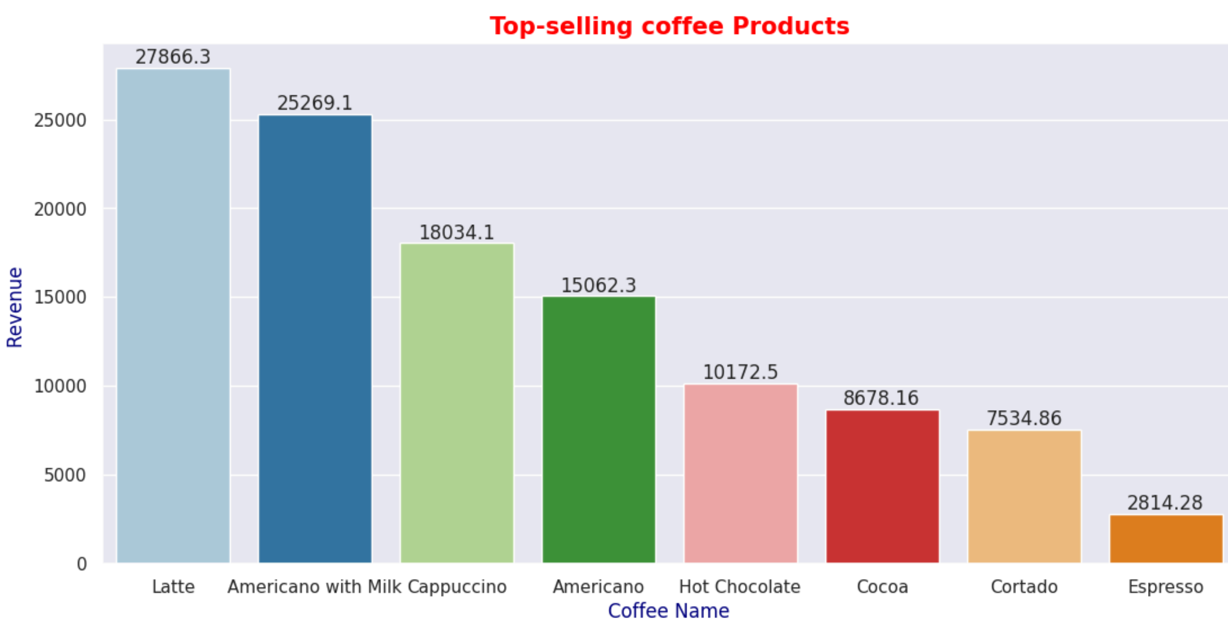
```
Top_selling_coffee = coffee.groupby("coffee_name")["money"].sum().reset_index(
Top_selling_coffee = Top_selling_coffee.sort_values(by = "money", ascending= F
Top_selling_coffee
```

Out[ ]:

	coffee_name	money
7	Latte	27866.30
1	Americano with Milk	25269.12
2	Cappuccino	18034.14
0	Americano	15062.26
6	Hot Chocolate	10172.46
3	Cocoa	8678.16
4	Cortado	7534.86
5	Espresso	2814.28

In [ ]:

```
plt.figure(figsize=(13,6))
sns.set_theme(style = "darkgrid")
ax = sns.barplot(x = "coffee_name", y = "money",
data = Top_selling_coffee, palette= "Paired")
for bars in ax.containers:
    ax.bar_label(bars)
plt.title("Top-selling coffee Products",
size =15, color="red", fontweight= "bold")
plt.xlabel("Coffee Name", size = 12, color = "Navy")
plt.ylabel("Revenue ", size = 12, color= "navy")
plt.show()
```



Latte (24.1%) and Americano with Milk (21.9%) together contribute nearly 46% of total coffee sales revenue.

Q2: Revenue Trends Over Time

In [ ]:

```
Monthly_Revenue= coffee.groupby("Month")["money"].sum().reset_index()
Monthly_Revenue = Monthly_Revenue.sort_values(by = "money", ascending= False)
Monthly_Revenue
```

Out[ ]:

	Month	money
7	Mar	17036.64
10	Oct	13891.16
3	Feb	13215.48
11	Sep	9988.64
8	May	9063.42
9	Nov	8590.54
2	Dec	8237.74
6	Jun	7758.76
1	Aug	7613.84
5	Jul	6915.94
0	Apr	6720.56
4	Jan	6398.86

In [ ]:

```
plt.figure(figsize= (10,6))
sns.set_theme(style= "darkgrid")
sns.lineplot(x = "Month", y = "money",
data = Monthly_Revenue)
plt.title("Revenue Trends Over Time ", size=15,
color= "red", fontweight= "bold")
plt.xlabel("Month", size=12, color="navy")
plt.ylabel("Revenue", size=12, color="navy")
plt.show()
```

Revenue increased by 166% from January to March, with March recording the highest sales and January the lowest.

Q3: Customer Buying Frequency

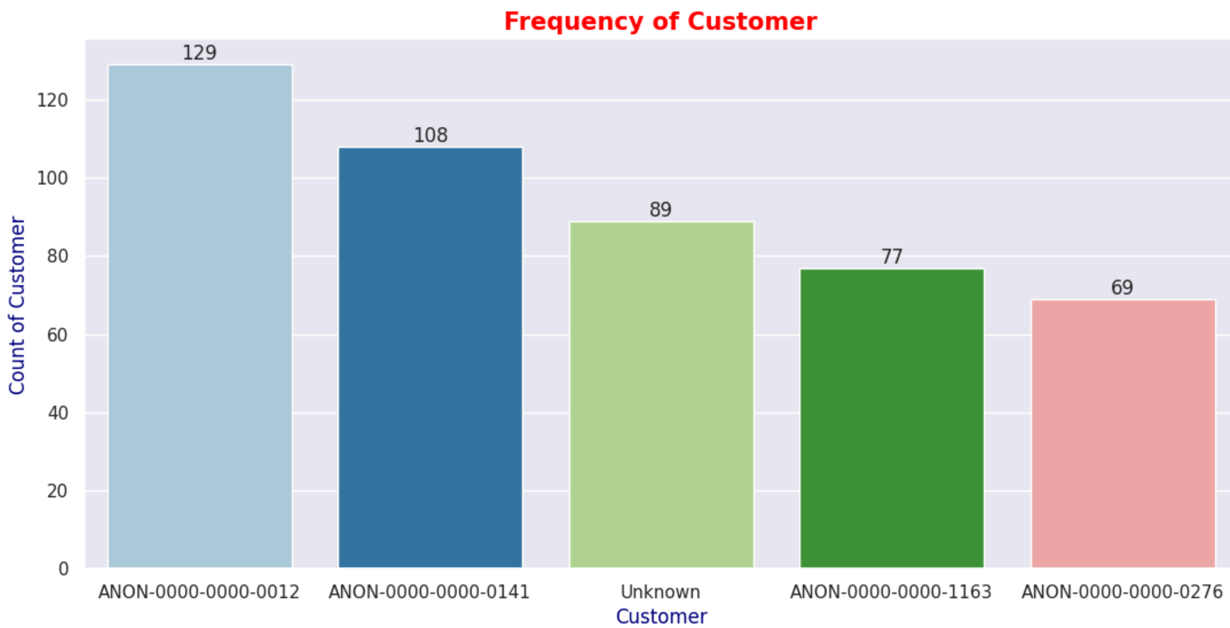
In [ ]:
```
Frequency_Customer = coffee["card"].value_counts().reset_index()[:5]
Frequency_Customer
```

Out[ ]:

	card	count
0	ANON-0000-0000-0012	129
1	ANON-0000-0000-0141	108
2	Unknown	89
3	ANON-0000-0000-1163	77
4	ANON-0000-0000-0276	69

In [ ]:
```
plt.figure(figsize=(13,6))
sns.set_theme(style= "darkgrid")
ax = sns.barplot(x = "card", y = "count",
data = Frequency_Customer, palette ="Paired")
for bars in ax.containers:
    ax.bar_label(bars)
plt.title("Frequency of Customer", size =15,
```

```
color= "red", fontweight="bold")
plt.xlabel("Customer", size =12,
color="navy")
plt.ylabel("Count of Customer", size=12,
color="navy")
plt.show()
```



Customer ANON-0000-0000-0012 made the highest number of purchases (129), followed by ANON-0000-0000-0141 (108). Unknown customers contributed 89 transactions, indicating a moderate share of unregistered buyers.

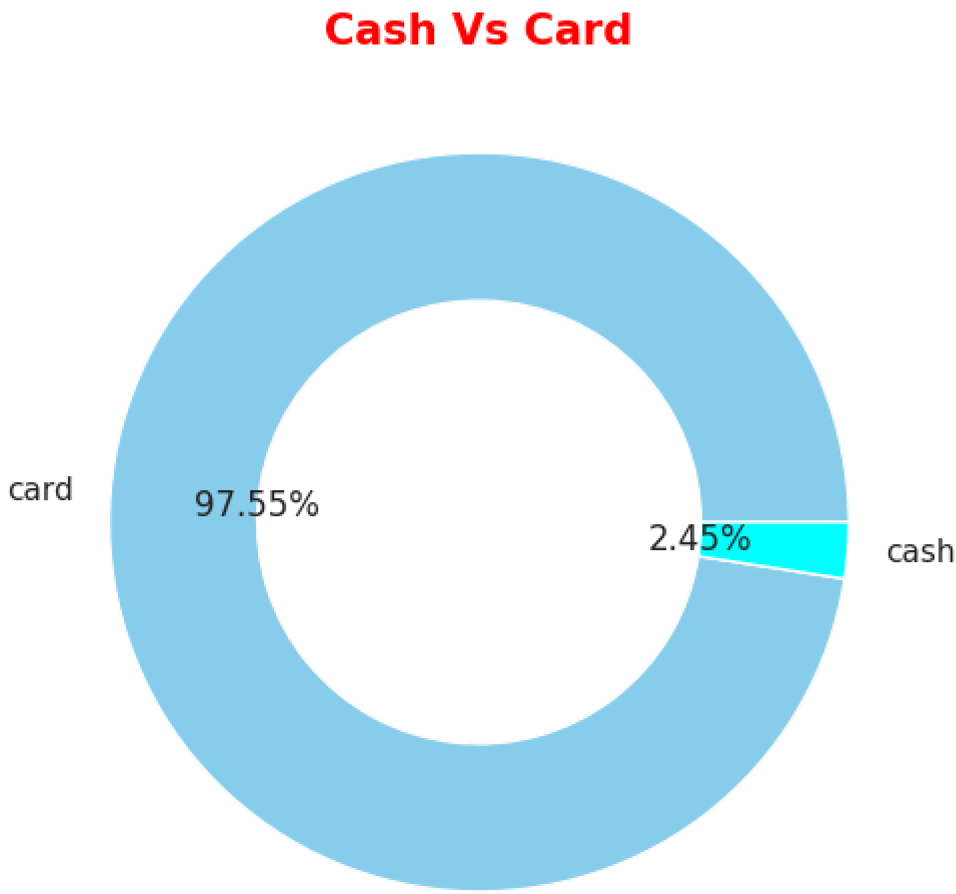
Q4: Cash vs Card Transactions

In [ ]:

```
Transactions= coffee["cash_type"].value_counts().reset_index()
x = Transactions["cash_type"]
y = Transactions["count"]
```

In [ ]:

```
plt.figure(figsize=(10,6))
plt.pie(y, labels=x, autopct="%0.2f%%",
wedgeprops={"width":0.4}, colors =["skyblue", "aqua"])
plt.title("Cash Vs Card", size =15,
color="red", fontweight="bold")
plt.show()
```



97.55% of customers used card payments for their transactions.

Q5: Peak Hours or weeks for Coffee Sales

In [ ]:
```
hourly_sales= coffee.groupby("Hours")["money"].sum().reset_index()
hourly_sales = hourly_sales.sort_values(by ="money", ascending=False)
hourly_sales
```

Out[ ]:

	Hours	money
4	10	10994.52
10	16	9221.60
5	11	8849.10
13	19	7966.96
11	17	7925.00
9	15	7789.02
6	12	7668.62
3	9	7429.28
8	14	7265.80
12	18	7235.60
7	13	7108.76
2	8	7017.88
15	21	6465.94
14	20	5656.92
16	22	3747.16
1	7	2940.02
0	6	149.40

In [ ]:

```
Weekly_sales = coffee.groupby("Week")["money"].sum().reset_index()
Weekly_sales = Weekly_sales.sort_values(by ="money", ascending=False)
Weekly_sales
```

Out[ ]:

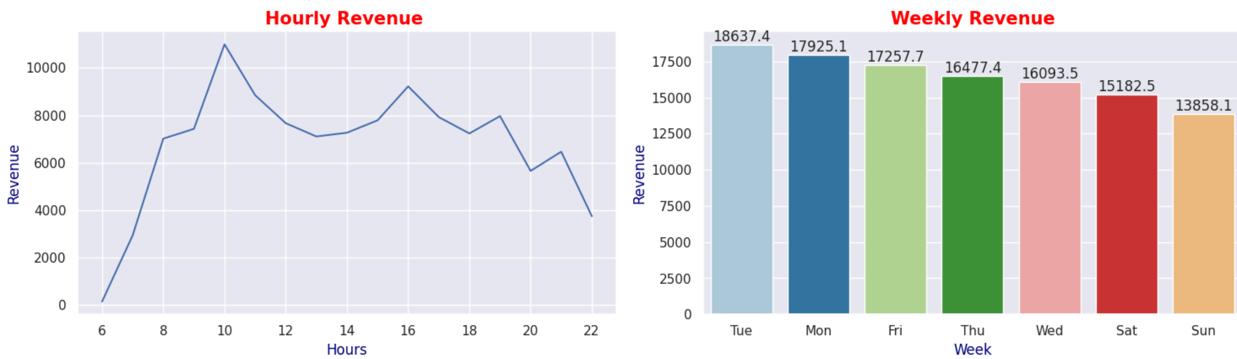
	Week	money
5	Tue	18637.38
1	Mon	17925.10
0	Fri	17257.66
4	Thu	16477.40
6	Wed	16093.46
2	Sat	15182.52
3	Sun	13858.06

In [ ]:

```
plt.figure(figsize=(15,8))

#Hourly Revenue
```

```
plt.subplot(2,2,1)
sns.lineplot(x = "Hours", y = "money",
data = hourly_sales)
plt.title("Hourly Revenue ", size =15,
color="red", fontweight="bold")
plt.xlabel("Hours", size=12,
color="navy")
plt.ylabel("Revenue", size=12,
color="navy")
#Weekly Revenue
plt.subplot(2,2,2)
ax = sns.barplot(x = "Week", y ="money",
data = Weekly_sales, palette="Paired")
for bars in ax.containers:
    ax.bar_label(bars)
plt.title("Weekly Revenue", size=15,
color="Red", fontweight="bold")
plt.xlabel("Week", size =12,
color="navy")
plt.ylabel("Revenue", size=12,
color ="navy")
plt.tight_layout()
plt.show()
```



Peak coffee sales occur at 10 AM and on Tuesdays, whereas the lowest sales happen at 6 AM and Sundays — highlighting strong weekday and morning demand patterns.

Consultation

Latte and Americano with Milk are the top-selling products, generating the highest revenue. Revenue increased by 166%, showing strong growth. Around 97.55% of customers prefer card payments, reflecting a shift toward digital transactions. Peak sales occur between 8–10 AM and on Tuesdays, suggesting the cafe should focus staff and marketing efforts during these periods.